

Scalability & Future Plan

Date	15 March 2024
Team ID	XXXXXX
Project Name	Smart Lender
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Local SQLite Database	Limits concurrent user write operations and cannot scale across multiple servers for a production deployment.	High
2	Lack of Model Explainability	The system outputs binary (Approved/Rejected) decisions, but does not tell the user or loan officer <i>why</i> the decision was made.	High
3	No API Endpoints	The application is tightly coupled to the web frontend, preventing external platforms (like mobile apps) from easily consuming the ML predictions.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Runs on the default single-threaded Flask development server.	Deploy using a production WSGI server like Gunicorn combined with an Nginx reverse proxy to handle concurrent user requests efficiently.
2	Data Storage	Stores data locally in an instance/smart_lender.db SQLite file.	Migrate to a robust, scalable relational database system like PostgreSQL or MySQL hosted on a cloud provider (e.g., AWS RDS).
3	Performance	The Machine Learning model is loaded directly into the web server's memory upon startup.	Decouple the web app from the ML model by deploying the model as a dedicated microservice (e.g., using FastAPI or a serverless function) to scale inference independently.
4	Security	Basic Flask session-based authentication with simple hashing.	Implement JWT (JSON Web Tokens) or OAuth 2.0 for API security, add rate-limiting to prevent abuse, and deploy behind HTTPS

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Model Explainability (SHAP/LIME)	Month 1	Increases transparency by showing officers the exact factors (e.g., income, credit history) that led to an approval or rejection.
Phase 2	RESTful API Architecture	Month 2	Allows the Smart Lender prediction engine to be easily integrated into mobile applications and third-party financial services.
Phase 3	Cloud Deployment & CI/CD	Month 3	Ensures high availability, automated testing, and seamless updates by deploying to AWS/GCP using Docker.